Optimization Methods

Seminar 07

Pre-colloquium Practice

Theory handout

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This handout is a compact review of what a colloquium ticket asks for: six definitions, two statements with proofs, two problems. The three sections below walk through part three; the cheat sheet is for parts one and two.

Notation.
$f : \mathbb{R}^2 \to \mathbb{R}$ is a quadratic form in two variables, Q is its Hessian, x^* its stationary point. $\mu = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$ when $Q \succ 0$. Gradient descent is $x_{k+1} = x_k - \alpha \nabla f(x_k)$ with a fixed $\alpha > 0$.

1 Classifying a stationary point

Definition. A point x^* is a **stationary point** of f when $\nabla f(x^*) = 0$. For a quadratic $f(x) = \frac{1}{2} x^\top H x + b^\top x$ this reads $Hx^* = -b$.

Theorem: Second-order test. Let f be twice continuously differentiable and $\nabla f(x^*) = 0$. If $\nabla^2 f(x^*) \succ 0$, then x^* is a strict local minimum. If $\nabla^2 f(x^*) \prec 0$, a strict local maximum. If $\nabla^2 f(x^*)$ is indefinite, a saddle point.

Proof. Taylor: $f(x^* + h) = f(x^*) + \frac{1}{2} h^\top \nabla^2 f(x^*) h + o(\|h\|^2)$. When $\nabla^2 f(x^*) \succ 0$ there is $m > 0$ with $h^\top \nabla^2 f(x^*) h \geq m \|h\|^2$, so $f(x^* + h) > f(x^*)$ for all small $h \neq 0$. The other cases swap the sign.

Why. For $f(x, y) = ax^2 + bxy + cy^2 + dx + ey$ the Hessian is the constant matrix $H = \begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}$. Its eigenvalues solve $\lambda^2 - \operatorname{tr}(H)\lambda + \det(H) = 0$; the signs follow from tr and det.

A1. Stationary point and type of a quadratic form

On the board

Statement. For $f(x, y) = 2x^2 + 2xy + 3y^2 - 8x - 2y$, find the stationary point and its type. Then find μ and L as the smallest and the largest eigenvalue of the Hessian.

Solution.

1. $\nabla f = (4x + 2y - 8,\ 2x + 6y - 2)$, $H = \begin{pmatrix} 4 & 2 \\ 2 & 6 \end{pmatrix}$. Setting the gradient to zero: $\begin{cases} 4x + 2y = 8 \\ 2x + 6y = 2 \end{cases}$. The determinant of H is 20, so the system has a unique solution.
2. Solve: $x^* = 2.2$, $y^* = -0.4$. Check: $\nabla f(x^*) = (0,\ 0)$ and $f(x^*) = -8.4$.
3. Eigenvalues of H solve $\det(H - \lambda I) = 0$, that is $\lambda^2 - 10\lambda + 20 = 0$. The roots are $\lambda_{\min} = 5 - \sqrt{5} \approx 2.7639$ and $\lambda_{\max} = 5 + \sqrt{5} \approx 7.2361$. Both are positive, so the stationary point is a strict minimum.
4. So $\mu \approx 2.7639$ and $L \approx 7.2361$. The condition number is $\kappa = L/\mu \approx 2.618$; gradient descent with the best fixed step contracts the error on the slow axis by $(\kappa - 1)/(\kappa + 1) \approx 0.4472$ per step.

Answer. Minimum at $(x^*, y^*) = (2.2, -0.4)$; $\mu \approx 2.7639$, $L \approx 7.2361$.

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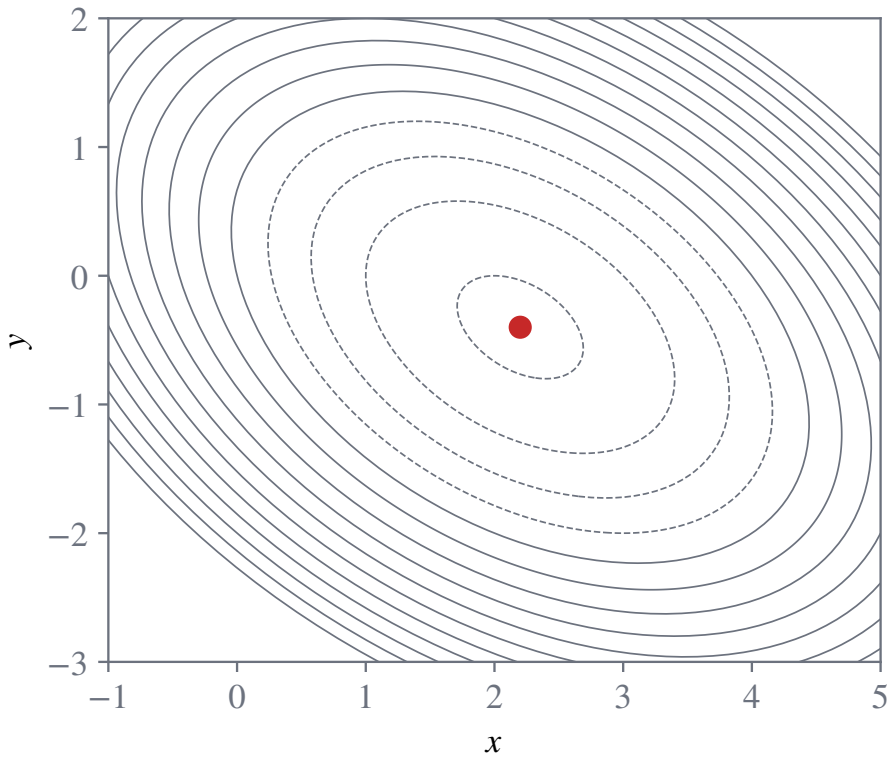


Figure. Level sets of f for problem A1 with the stationary point marked.

A2. Stationary point of another quadratic

On your own

Statement. For $f(x, y) = 4x^2 - 2xy + y^2 - 6x + 4y$, find the stationary point, its type, and μ and L.

Solution.

1. $\nabla f = (8x - 2y - 6,\ -2x + 2y + 4)$, $H = \begin{pmatrix} 8 & -2 \\ -2 & 2 \end{pmatrix}$. Setting the gradient to zero: $\begin{cases} 8x - 2y = 6 \\ -2x + 2y = -4 \end{cases}$. $\det H = 12 > 0$, so a unique solution.
2. Solve: $x^* = 1/3 \approx 0.333$, $y^* = -5/3 \approx -1.667$. Check: $\nabla f(x^*) = (0,\ 0)$ and $f(x^*) = -13/3 \approx -4.333$.
3. Eigenvalues of H: $\lambda^2 - 10\lambda + 12 = 0$, so $\lambda \approx 1.3944$ and $\lambda \approx 8.6056$. Both positive, the point is a strict minimum.
4. So $\mu \approx 1.3944$ and $L \approx 8.6056$. The condition number $\kappa \approx 6.17$; gradient descent at the best fixed step contracts by $(\kappa - 1)/(\kappa + 1) \approx 0.721$ per step.

Answer. Minimum at $(1/3, -5/3)$; $\mu \approx 1.3944$, $L \approx 8.6056$.

Trap. A gradient that is a row of zeros means a stationary point was found, not that it is a minimum. The second-order test is the step that cannot be skipped.

A3. A quadratic form with a saddle point

On the board

Statement. For $f(x, y) = x^2 - 4xy + y^2 - 6x + 6y$, find the stationary point and show that it is a saddle. What are μ and L here?

Solution.

1. $\nabla f = (2x - 4y - 6,\ -4x + 2y + 6)$, $H = \begin{pmatrix} 2 & -4 \\ -4 & 2 \end{pmatrix}$. The system $\nabla f = 0$ reads $\begin{cases} 2x - 4y = 6 \\ -4x + 2y = -6 \end{cases}$. $\det H = -12 < 0$: negative, so the eigenvalues have opposite signs.
2. Solve: $x^* = 1$, $y^* = -1$. Check: $\nabla f(x^*) = (0,\ 0)$ and $f(x^*) = -6$.
3. Eigenvalues of H: $\lambda^2 - 4\lambda - 12 = 0$. Roots: $\lambda_{\min} = -2$ and $\lambda_{\max} = 6$. Different signs, so the stationary point is a saddle: f grows along one eigenvector and shrinks along the other.
4. With an indefinite Hessian the constants μ and L of strong convexity and L-smoothness are not defined: f is neither convex nor L-smooth near a saddle, and the condition number does not apply.

Answer. Saddle at $(1, -1)$; eigenvalues -2 and 6. Neither μ nor L is defined.

A4. A quadratic form with a large condition number

On your own

Statement. For $f(x, y) = x^2 + 10y^2 - 2x - 20y$, find the stationary point, its type, μ, L and the condition number $\kappa = L/\mu$.

Solution.

1. $\nabla f = (2x - 2,\ 20y - 20)$, $H = \begin{pmatrix} 2 & 0 \\ 0 & 20 \end{pmatrix}$. Setting the gradient to zero: $\begin{cases} 2x = 2 \\ 20y = 20 \end{cases}$. $\det H = 40 > 0$, a unique solution.
2. Solve: $x^* = 1$, $y^* = 1$. Check: $\nabla f(x^*) = (0,\ 0)$ and $f(x^*) = -11$.
3. Eigenvalues of H: $\lambda^2 - 22\lambda + 40 = 0$. Roots: $\lambda = 2$ and $\lambda = 20$. Both positive, so the point is a strict minimum.
4. So $\mu = 2$, $L = 20$, $\kappa = L/\mu = 10$. Gradient descent at the best fixed step contracts the slow axis by $(\kappa - 1)/(\kappa + 1) = 9/11 \approx 0.8182$ per step: a step shrinks the error by less than a fifth of what the fast axis would allow.

Answer. Minimum at $(1, 1)$; $\mu = 2$, $L = 20$, $\kappa = 10$.

2 Strong convexity and smoothness

> **Definition.** f is **μ-strongly convex** when $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2$ for every x, y; it is **L-smooth** when $\|\nabla f(y) - \nabla f(x)\| \leq L\|y - x\|$ for every x, y.

For a twice differentiable f both are equivalent to $\mu I \preceq \nabla^2 f(x) \preceq LI$ at every x. For a positive definite quadratic $f(x) = \frac{1}{2}x^\top Qx + b^\top x$ the Hessian is Q, so $\mu = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$ at every point.

Example. For $f(x, y) = 2x^2 + 2xy + 3y^2$ the Hessian is $\begin{pmatrix} 4 & 2 \\ 2 & 6 \end{pmatrix}$; eigenvalues $\mu \approx 2.7639$, $L \approx 7.2361$, condition number $\kappa = L/\mu \approx 2.618$.

3 Gradient descent on a quadratic

Definition. **Gradient descent** at a fixed step $\alpha > 0$ is $x_{k+1} = x_k - \alpha \nabla f(x_k)$.

Theorem: Safe step. For $f(x) = \frac{1}{2}x^\top Qx$ with $Q \succ 0$, the iterates satisfy $x_{k+1} = (I - \alpha Q)x_k$, and $x_k \to 0$ iff $0 < \alpha < 2/L$. At $\alpha = 1/L$ the error on the direction of $\lambda_{\max}$ vanishes in one step; on $\lambda_{\min}$ it contracts by $1 - \mu/L$ per step.

Proof. In the eigenbasis of Q the iteration is $e_{k+1,i} = (1 - \alpha\lambda_i)e_{k,i}$, so $|e_{k,i}| = |1 - \alpha\lambda_i|^k |e_{0,i}|$. The whole sequence tends to zero iff every $|1 - \alpha\lambda_i| < 1$, iff $0 < \alpha < 2/\lambda_i$ for every i, iff $0 < \alpha < 2/L$.

B1. Three steps of gradient descent on a quadratic

On the board

Statement. Let $f(x, y) = x^2 - xy + y^2$. From $x_0 = (6, 0)$ take three steps of gradient descent with $\alpha = 1/3$. Report the gradients, the iterates x_1, x_2, x_3 and the values of f. Find L and check that $\alpha < 2/L$.

Solution.

1. $Q = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, so $\nabla f(x) = Qx$. At $x_0 = (6, 0)$: $\nabla f(x_0) = (12, -6)$. The step is $x_1 = x_0 - \alpha \nabla f(x_0) = (2, 2)$ and $f(x_1) = 4$.
2. At x_1: $\nabla f(x_1) = (2, 2)$, $x_2 = (4/3, 4/3) \approx (1.333, 1.333)$, $f(x_2) = 16/9 \approx 1.778$. At x_2: $\nabla f(x_2) = (4/3, 4/3)$, $x_3 = (8/9, 8/9) \approx (0.889, 0.889)$, $f(x_3) = 64/81 \approx 0.790$. The values fall from 36 to ≈ 0.79.
3. Eigenvalues of Q: $\lambda_{\min} = 1$ and $\lambda_{\max} = 3$, so $L = 3$ and $2/L \approx 0.667$. Here $\alpha = 1/3 < 2/L$: the step is safe.
4. The contraction factor along the slowest axis is $1 - \alpha\mu = 2/3 \approx 0.667$; the ratio $\|x_3\|/\|x_0\| \approx 0.209$ matches it to within the two-axis mixing. The iterates head to $(0, 0)$.

Answer. $f(x_1) = 4$, $f(x_3) \approx 0.790$; $L = 3$, $\alpha < 2/L$ holds.

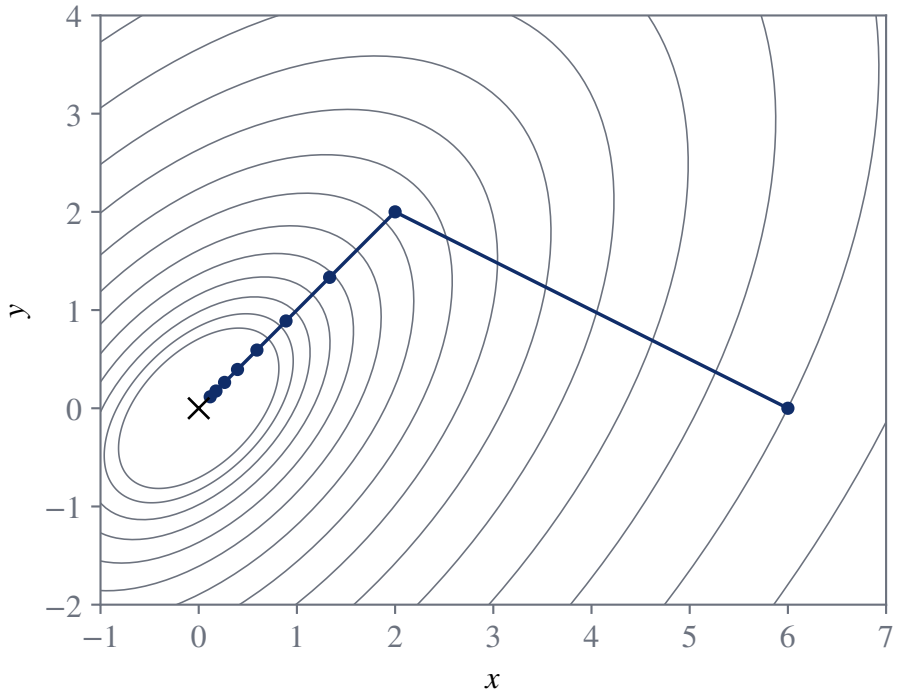


Figure. Three gradient-descent steps on problem B1 on the contour plot.

B2. Three steps on a diagonal quadratic

On your own

Statement. Let $f(x, y) = 3x^2 + 2y^2$. From $x_0 = (4, -3)$ take three steps of gradient descent with $\alpha = 1/6$. Report $f(x_1)$, $f(x_3)$, L and whether the step is safe.

Solution.

1. $Q = \begin{pmatrix} 6 & 0 \\ 0 & 4 \end{pmatrix}$. At $x_0 = (4, -3)$: $\nabla f = (24, -12)$. Step: $x_1 = x_0 - \alpha \nabla f = (0, -1)$, $f(x_1) = 2$. The first coordinate is wiped out in one step: $1 - 2\alpha a = 1 - 6/6 = 0$.
2. At x_1: $\nabla f = (0, -4)$, $x_2 = (0, -1/3) \approx (0, -0.333)$, $f(x_2) = 2/9 \approx 0.222$. At x_2: $\nabla f = (0, -4/3)$, $x_3 = (0, -1/9) \approx (0, -0.111)$, $f(x_3) \approx 0.0247$. The slow axis contracts by $1 - 2\alpha c = 1/3$ per step.
3. Eigenvalues of Q: $\mu = 4$ and $L = 6$, so $2/L = 1/3 \approx 0.333$. Here $\alpha = 1/6 < 2/L$, the step is safe. Actually $\alpha = 1/L$: the border between the fast axis meeting the floor and the slow axis contracting.
4. The ratio $\|x_3\|/\|x_0\| \approx 0.0222$ matches $(1/3)^3 \cdot |y_0|/\|x_0\| = 3/135 \approx 0.0222$: the slow axis set the rate for all three steps.

Answer. $f(x_1) = 2$, $f(x_3) \approx 0.0247$; $L = 6$, $\alpha < 2/L$ holds.

Trap. The border $\alpha = 2/L$ is not safe: the direction of $\lambda_{\max}$ oscillates with constant amplitude, so the iterates stop moving but do not converge.

B3. Three steps on the border $\alpha = 2/L$

On the board

Statement. Let $f(x, y) = x^2 + y^2$. From $x_0 = (2, 1)$ take three steps of gradient descent with $\alpha = 1$ (which equals $2/L$). Report x_1, x_2, x_3, the values of f, and describe how the iterates behave.

Solution.

1. $Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, $L = 2$, $2/L = 1$. Here $\alpha = 1 = 2/L$ exactly. On each coordinate the step multiplier is $1 - \alpha L = -1$.
2. At $x_0 = (2, 1)$: $\nabla f(x_0) = (4, 2)$, $x_1 = x_0 - \alpha \nabla f(x_0) = (-2, -1)$, $f(x_1) = 5$. The iterate flipped sign; the magnitude is unchanged.
3. $x_2 = (2, 1)$, $f(x_2) = 5$; $x_3 = (-2, -1)$, $f(x_3) = 5$. The sequence oscillates between x_0 and $-x_0$; $\|x_k\| = \sqrt{5}$ at every step and f stays at 5.
4. This is the border case: $|1 - \alpha L| = 1$. The step is not safe — the method does not diverge, but it does not converge either; a strict inequality $\alpha < 2/L$ is needed.

Answer. $f(x_1) = f(x_2) = f(x_3) = 5$; $\|x_3\|/\|x_0\| = 1$. The iterates oscillate between x_0 and $-x_0$ with constant amplitude.

B4. Three steps when $\alpha > 2/L$

On your own

Statement. Let $f(x, y) = x^2 + y^2$ again. From $x_0 = (1, -1)$ take three steps of gradient descent with $\alpha = 3/2$ (which is above $2/L$). Report $f(x_1)$, $f(x_3)$, and the per-step growth factor $|1 - \alpha L|$ on each coordinate.

Solution.

1. $L = 2$, $2/L = 1$, so $\alpha = 3/2 > 2/L$. The per-coordinate multiplier is $1 - \alpha L = -2$: its absolute value is $|1 - \alpha L| = 2 > 1$.
2. At $x_0 = (1, -1)$: $\nabla f(x_0) = (2, -2)$, $x_1 = x_0 - \alpha \nabla f(x_0) = (-2, 2)$, $f(x_1) = 8$. Each coordinate changed sign and doubled in absolute value.
3. $x_2 = (4, -4)$, $f(x_2) = 32$; $x_3 = (-8, 8)$, $f(x_3) = 128$. The coordinates grow by 2 each step, f grows by 4 each step.
4. The ratio $\|x_3\|/\|x_0\| = 8 = |1 - \alpha L|^3$. The method diverges; nothing of the quadratic matters here except the top eigenvalue of Q, that is L.

Answer. $f(x_1) = 8$, $f(x_3) = 128$; $|1 - \alpha L| = 2$ per coordinate per step, so the method diverges.